

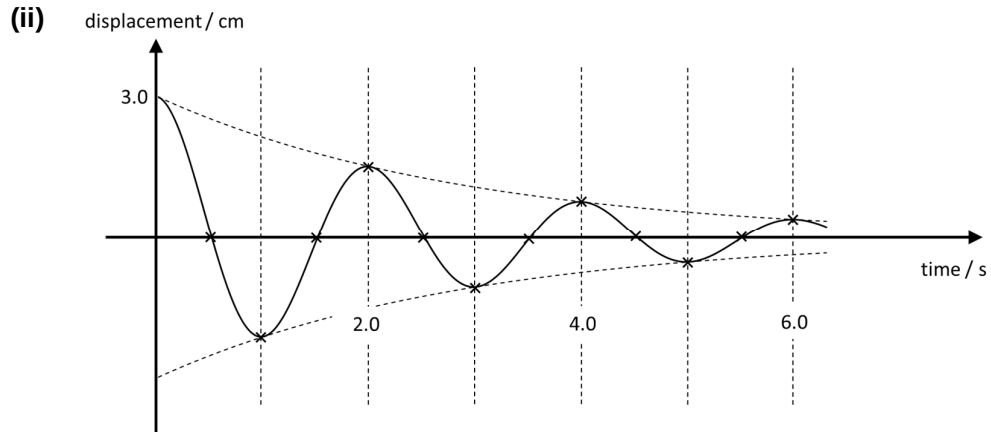
- 3 (a) A damped oscillation refers to the oscillator's energy being dissipated Or resistive/dissipative forces are present OR amplitude of oscillation decreases over time.

(b) (i) $a = -\omega^2 x$

$$|a_{\max}| = \left| -\left(\frac{2\pi}{T}\right)^2 3.0 \times 10^{-2} \right| \quad [\text{M1}]$$

$$= 0.296 \text{ m s}^{-2} \quad [\text{A1}]$$

Accept $a_0 = \omega^2 x_0$



[B1] light damping with decreasing displacement.

[B1] initial displacement of 3.0 cm & period shown as 2.0 s.

- (c) physical phenomenon: resonance [B1]

condition: driving frequency equals to the natural frequency of the spring-mass system [B1]

- (d) Critical damping is required for a car suspension system. [B1]

This is necessary to stop the car from oscillating as quickly as possible that could otherwise cause driver to lose control of the car/ cause excessive damage to car/ cause excessive discomfort to passengers/returns to equilibrium as quick as possible such that it ready for next hump on the road. [B1]